

Understanding Analysis, 2nd Edition

Chapter 5 - Section 5.3 - The Mean Value Theorems

1. Recall from Exercise 4.4.9 that a function $f : A \rightarrow \mathbf{R}$ is Lipschitz on A if there exists an $M > 0$ such that

$$\left| \frac{f(x) - f(y)}{x - y} \right| \leq M$$

for all $x \neq y$ in A .

- (a) Show that if f is differentiable on a closed interval $[a, b]$ and if f' is continuous on $[a, b]$, then f is Lipschitz on $[a, b]$.
- (b) Review the definition of a contractive function in Exercise 4.3.11. If we add the assumption that $|f'(x)| < 1$ on $[a, b]$, does it follow that f is contractive on this set?

Solution:

- (a) Continuous function preserves compactness (Theorem 4.4.1) and compact sets are bounded. Since f' is continuous on $[a, b]$, it's bounded. Thus there exists $M > 0$ such that $|f'(c)| \leq M$ for all $c \in [a, b]$.

Now, by the Mean Value Theorem, for any $x \neq y$ in $[a, b]$, there exists t in $[a, b]$ such that

$$\frac{f(x) - f(y)}{x - y} = f'(t)$$

Therefore

$$\left| \frac{f(x) - f(y)}{x - y} \right| = |f'(t)| \leq M$$

as required.

- (b) f is contractive if there exists a constant $0 \leq c < 1$ such that $|f(x) - f(y)| \leq c|x - y|$ for all $x, y \in \mathbf{A}$.

The answer is yes. From part (a), we have

$$|f(x) - f(y)| = |f'(t)||x - y|$$

And we have $|f'(t)| < 1$.

But we need to choose a single constant c (whereas $|f'(t)|$ depends on x and y). From part (a), f' is continuous on a compact set $[a, b]$ so it attains a maximum.

Choose $c = \max_{x \in [a,b]} |f'(x)|$. Since $|f'(x)| < 1$ on $[a, b]$ from the problem's assumption, the contractive condition is satisfied.

2. Let f be differentiable on an interval A . If $f'(x) \neq 0$ on A , show that f is one-to-one on A . Provide an example to show that the converse statement need not be true.

Solution: We prove the contrapositive. Suppose f is not one-to-one, then there exists $a < b$ in A such that $f(a) = f(b)$. By the Mean Value Theorem, there exists $c \in A$ such that

$$\frac{f(b) - f(a)}{b - a} = 0 = f'(c)$$

so f' cannot be non-zero on all of A .

Counterexample to the converse: x^2 on $[0, 1]$. It's one-to-one (because we exclude negative domain) with $f'(0) = 0$.

Alternatively, a cleaner counterexample that doesn't involve one-sided derivative is x^3 .

3. Let h be a differentiable function defined on the interval $[0, 3]$, and assume that $h(0) = 1$, $h(1) = 2$, and $h(3) = 2$.
- (a) Argue that there exists a point $d \in [0, 3]$ where $h(d) = d$.
 - (b) Argue that at some point c we have $h'(c) = 1/3$.
 - (c) Argue that $h'(x) = 1/4$ at some point in the domain.

Solution:

- (a) The fixed point d is not necessarily in $[0, 1]$, as we can draw a straight line between $h(0)$ and $h(1)$ and there'd be no fixed point. So it must be in $[1, 3]$.

The proof is basically the same as Exercise 4.5.7. Define $g(x) = h(x) - x$ on $[1, 3]$. If there exists a fixed point d then it must be that $g(d) = 0$. We have $g(1) = 1$ and $g(3) = -1$. g is continuous, so we can use IVT to conclude that there exists such a d .

- (b) The slope between $x = 0$ and $x = 3$ is $\frac{h(3)-h(0)}{3-0} = 1/3$. By the Mean Value Theorem, there exists a $c \in (0, 3)$ such that $h'(c) = 1/3$.

- (c) Applying the same argument as part (b) to $x = 1$ and $x = 3$, we can conclude that there exists $d \in (1, 3)$ such that $h'(d) = 0$. Now we have

$$(h'(d) = 0) < 1/4 < (h'(c) = 1/3)$$

Apply Darboux Theorem to conclude that there exists $x \in (0, 3)$ such that $h'(x) = 1/4$.

4. Let f be differentiable on an interval A containing zero, and assume (x_n) is a sequence in A with $(x_n) \rightarrow 0$ and $x_n \neq 0$.

- (a) If $f(x_n) = 0$ for all $n \in \mathbf{N}$, show $f(0) = 0$ and $f'(0) = 0$.
(b) Add the assumption that f is twice-differentiable at zero and show that $f''(0) = 0$ as well.

Solution:

- (a) $f(x_n)$ is the zero sequence, so we have $(x_n) \rightarrow 0 \implies f(x_n) \rightarrow 0$. Since f is differentiable on A , it's continuous on A , so we have $f(0) = 0$.

Note that it's enough to prove this for the one sequence (x_n) instead of *any* sequence that converges to zero, because we're given that f is differentiable, so its limit is well defined and any sequence that converges to zero has the same functional limit.

For $f'(0)$, we compute it from the definition. Using the fact that $f(x_n) = 0$

$$f'(0) = \lim_{x_n \rightarrow 0} \frac{f(x_n) - f(0)}{x_n - 0} = 0/x_n = 0$$

Again, it's enough to prove this for a single sequence (x_n) because we know in advance that f is differentiable. Any sequence that converges to zero must produce the same derivative.

- (b) TODO.

5. (a) Supply the details for the proof of Cauchy's Generalized Mean Value Theorem (Theorem 5.3.5).
- (b) Give a graphical interpretation of the Generalized Mean Value Theorem analogous to the one given for the Mean Value Theorem at the beginning of Section 5.3. (Consider f and g as parametric equations for a curve).

Solution:

- (a) Define

$$h(x) = [f(b) - f(a)]g(x) - [g(b) - g(a)]f(x)$$

We have $h(a) = h(b)$, so we can apply Rolle's Theorem to conclude that there exists $c \in (a, b)$ such that

$$h'(c) = 0 = [f(b) - f(a)]g'(c) - [g(b) - g(a)]f'(c)$$

And rearranging gives $[g(b) - g(a)]f'(c) = [f(b) - f(a)]g'(c)$ as required. If $g'(c) \neq 0$, we can also write it as $\frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}$.

- (b) Consider the curve defined by the parametric equation $(g(t), f(t))$. Note the ordering, g for the x -axis and f for the y -axis.

A line from $(g(a), f(a))$ to $(g(b), f(b))$ has the slope $\frac{f(b)-f(a)}{g(b)-g(a)}$. This is the rhs of the Theorem (in its fractional form).

The derivative of the curve at a point c is given by

$$\begin{aligned} \lim_{t \rightarrow c} \frac{f(t) - f(c)}{g(t) - g(c)} &= \lim_{t \rightarrow c} \frac{[f(t) - f(c)] * \frac{1}{t-c}}{[g(t) - g(c)] * \frac{1}{t-c}} \\ &= \frac{f'(c)}{g'(c)} \end{aligned}$$

and this is the lhs of the Theorem.

In conclusion, given a line from $(g(a), f(a))$ to $(g(b), f(b))$, the Generalized Mean Value Theorem says that there exists a tangent to the curve at $(g(c), f(c))$ with the same slope (i.e. parallel) to the line, provided $g'(c) \neq 0$.

The regular Mean Value Theorem is a special case where $g(t) = t$.